



Development of biopolymer composites using lignin: A sustainable technology for fostering a green transition in the construction sector

Barney H. Miao^{a,1,*}, Robert J. Headrick^b, Zhiye Li^{a,2}, Leonardo Spanu^b, David J. Loftus^{c,*}, Michael D. Lepech^{a,3}

^a Department of Civil and Environmental Engineering, Stanford University, 473 Via Ortega, Stanford, CA 94305, United States of America

^b Shell International Exploration & Production Inc, Houston, TX 77082, United States of America

^c NASA Ames Research Center, De France Ave, Moffett Field, CA 94035, United States of America

ARTICLE INFO

Keywords:

Lignin
Sustainable construction
Biopolymer-bound soil composite
Sustainable material development
Lignin-based material design
Green transition

ABSTRACT

Developing sustainable construction materials is important to help reduce the anthropogenic impacts of the construction industry. Currently, the production of concrete accounts for 8 % of global carbon emissions. Therefore, alternatives to concrete must be developed, to reduce its use in the future. New construction materials will help to facilitate a green transition as envisioned in global climate initiatives. Materials such as lignin are ideal, as they can be implemented with little additional cost to manufacture construction materials. We introduce a novel material, lignin-based biopolymer-bound soil composite (BSC), which is similar to other BSCs using other types of biopolymers. In addition, a design methodology is presented, which allows the manufacture of lignin-based BSCs with tailored characteristics. Two kinds of lignin — hydrolysis lignin and alkali lignin — were investigated, with five mix designs developed for each type of lignin. The lignin-based BSCs were found to have compressive strength ranging from 1.6–8.1 MPa, which allows them to be implemented in non-structural construction applications. Ultimate compressive strength, density, and other parameters were measured, leading to the development of design relationships for lignin-based BSC. The design relationships presented in this study will help introduce lignin-based BSC as a sustainable form of construction.

1. Introduction and motivation

Biopolymer-bound soil composite (BSC) shows promise as a potential substitute for concrete, bricks, and other rigid construction materials. Unlike concrete, which uses portland cement, BSC uses various biopolymers, such as proteins, starches, and others to solidify soil (Khatami and O'Kelly, 2013; Chang and Cho, 2014; Chang et al., 2015a; Ayeldeen et al., 2016; Qureshi et al., 2017; Dove et al., 2016; Chang et al., 2015b; Muguda et al., 2017). This strategy is intended to reduce carbon emissions associated with concrete production, which accounts for approximately 8% of global carbon emissions (Albuhairi and Di Sarno, 2022; Hasanbeigi et al., 2012; Bribián et al., 2011; De Schepper et al., 2014). BSC requires aggregate, a biopolymer and a solvent that is capable of dissolving the biopolymer. Removal of the solvent by evaporation or other forms of desiccation allows BSC to develop its strength. BSC has strength similar to that of conventional construction materials, but a much lower life cycle impact (Miao et al.,

2024b,a). The development of BSC is part of larger push towards developing more sustainable construction materials to help reduce the carbon footprint of the built environment (Lou et al., 2023b; Wang et al., 2022).

In this paper, we introduce a novel form of BSC, which uses lignin as the biopolymer. Lignin is structurally complex and is composed of a cross-linked three-dimensional structure made of p-hydroxyphenyl, guaiacyl, and syringyl units (Grabber, 2005; Lu et al., 2020). The proportions of these units vary with the source of lignin. Guaiacyl units and p-hydroxyphenyl units predominate in softwood lignin, and guaiacyl units and syringyl units predominate in hardwood lignin (Katahira et al., 2018). Chemical groups such as hydroxyl, carbonyl, and carboxyl groups also vary based on the source of lignin (Figueiredo et al., 2018; Brunow and Lundquist, 2010).

Lignin is a waste product from paper and pulp production and from production of cellulosic bio-fuels (Bajwa et al., 2019). Materials originating from untapped industrial waste streams are attractive because

* Corresponding authors.

E-mail addresses: barneym@stanford.edu (B.H. Miao), david.j.loftus@nasa.gov (D.J. Loftus).

¹ PhD Candidate.

² Research Engineer.

³ Professor.

they reduce the need for additional raw materials. Currently, around 50–70 million tons of lignin are produced worldwide (Collins et al., 2019; Mankar et al., 2022). With initiatives such as the Renewable Fuel Standard (RFS), the annual production of lignin is forecast to increase 5-fold by 2030 due to an anticipated increase in biofuel production (Bajwa et al., 2019). Lignin is typically considered to be a low-value waste product, where 98%–99% is combusted as fuel. Only 1%–2% of lignin is used in producing higher-value products such as syngas, carbon fibers, and synthetic phenolic compounds (Luo and Abu-Omar, 2017; Demuner et al., 2021; Boarino and Klok, 2023). Most lignin (> 60 % carbon), however, is burned onsite, in the open, leading to significant emissions of carbon into the atmosphere (Pflieger and Kroflič, 2017). A more ideal use for lignin would be to incorporate it into construction materials, such as bricks, which will help prevent the generation of harmful pollutants and sequester carbon in the long term (Miao et al., 2024b).

Prior studies of construction applications of lignin have been focused on its potential in soil stabilization rather than in forming structural components (Kong et al., 2019; Canakci et al., 2015; Zhang et al., 2020, 2018; Liu et al., 2020). Kong et al. demonstrated the potential of using lignin as a substitute for traditional stabilizers (cement, lime, etc.). Conventional soil stabilizers raise soil pH, leading to adverse effects on the environment (groundwater and surrounding vegetation) (Latifi et al., 2017). According to ASTM C55 and ASTM C129, the minimum net area compressive strength for load-bearing and non-structural concrete masonry units are 17.2 MPa and 3.45 MPa. Therefore, lignin-based BSC should have a compressive strength of at least 3.45 MPa to be considered viable for non-structural construction (non-load bearing partition walls, non-load bearing masonry units, etc.).

In this paper we introduce a novel version of BSC which uses lignin as the biopolymer binder. We use the term “lignin-based BSC” to refer to this new material. To characterize this material we fabricated a series of test specimens made from different mixtures of sand and lignin, using DMSO to dissolve the lignin. We investigated two different types of lignin, hydrolysis lignin and alkali lignin. We chose these types of lignin because they come from different industrial processes. Alkali lignin is a byproduct of the paper-making industry. Hydrolysis lignin is a byproduct of the bio-ethanol industry. While not as abundant as other types of lignin, presently, production of hydrolysis lignin is expected to grow, substantially, along with the growth of the bio-ethanol industry. To determine the compressive strength of these materials we performed unconfined, uni-axial compressive strength testing. Lastly, we developed a set of design relationships, which pave the way for use of this material in select construction applications.

2. Mixture design theory

Biopolymer-bound soil composites are comprised of densely packed aggregates, with the inter-particle voids filled by a biopolymer binder. The compressive strength is dependent on the strength of the aggregate, the interfacial bond, and the biopolymer matrix. Previous studies have speculated that the main determinant of BSC's compressive strength is due to the soil particles taking the load, with the biopolymer matrix providing shear resistance (Roedel, 2017; Rosa et al., 2020). Strategies to increase the strength of BSC include selecting biopolymers with greater mechanical strength and increasing the strength of the interfacial bond between the biopolymer and the aggregate. To determine the optimal mix, the concept of dry bulk soil density is used, where this term refers to the density of the aggregate in the finished (dried) BSC. Table 1 shows the proposed mix design framework. Each step is further discussed in this section. The design methodology of lignin-based BSC closely follows prior BSCs, with the main difference being the use of a non-water solvent — DMSO — to dissolve the biopolymer into solution (Rosa et al., 2020).

1. Find a solvent that can dissolve the biopolymer

The lignins investigated in this study are hydrolysis lignin (HL) and alkali lignin (AL) (Sigma Aldrich, 2023). Unlike previously explored biopolymers, hydrolysis lignin and alkali lignin are insoluble in water under normal conditions ($pH \approx 7.0$). This is an important issue since the biopolymer cannot bond to the aggregate if it does not start out fully dissolved. Therefore, we used DMSO as an alternate solvent, in order to dissolve lignin at high concentrations (> 20 %). It is worthwhile to mention that BSCs made from hydrophobic biopolymers will likely have greater durability in wet environments, as compared to BSCs made with hydrophilic biopolymers. BSCs made with hydrophobic biopolymers, such as lignin, may be more effective for outdoor applications without the need for waterproof coatings. Prior studies indicate that a variety of organic solvents are effective in dissolving lignin, including butyl carbitol, dioxane, ethylene glycol, and dimethylsulfoxide (DMSO) (Sameni et al., 2017; Dastpak et al., 2020; Sigma Aldrich, 2023). For this study, we selected dimethylsulfoxide (DMSO), since it is low cost and poses few safety concerns.

2. Determine the maximum amount of biopolymer that can be dissolved into solution

The maximum amount of biopolymer that can be practically dissolved into solution is limited by the workability limit for the biopolymer, which is defined as the maximum amount of biopolymer (lignin) that can be dissolved into solution such that it does not undergo glass-like kinetic arrest and is still practically workable (Rosa et al., 2020). Even though prior literature on the workability limit of lignin is not available, there is a rough estimate based on the workability limit of bovine serum albumin (BSA). According to Brownsey et al. the upper limit for BSA solubility, $(\%w/w_{BP})_{max}$, is 55% (Brownsey et al., 2003). However, with different equipment, Rosa et al. was only able to experimentally achieve 48% solubility (Rosa et al., 2020). We found that the upper limits of solubility for the two types of lignin are 47% and 58% for hydrolysis lignin and alkali lignin, in DMSO, respectively. The following equations define the biopolymer concentration, $\%w/w_{BP}$, of the solution:

$$(\%w/w_{BP}) = \frac{m_{BP}}{m_{BP} + m_{solvent}} = \frac{m_{BP}}{m_{BPSol.}} \quad (1)$$

where $m_{BPSol.}$ is the mass of the solution, which is equal to the combined mass of the biopolymer, m_{BP} , and the solvent, $m_{solvent}$.

3. Find the density of the biopolymer solution $\rho_{BPSol.}$

The density of the biopolymer solution changes with the concentration of biopolymer in the solution. Eq. (2) will be used to find the density of the biopolymer solution. For this equation, it is assumed that the temperature of DMSO and the resulting solution is at 25 °C and that the partial specific volume of the two lignins are the same. Either of these assumptions will lead to negligible effects on the results produced. The density of the biopolymer solution is given by the following equation:

$$\begin{aligned} \rho_{BPSol.} &= \frac{1}{\sum m_i \bar{v}_i} \\ &= \frac{1}{\%w/w_{BP} * a + (1 - \%w/w_{BP}) * b} \\ a &= 0.769 \text{ cm}^3/\text{g}, b = 0.9128 \text{ cm}^3/\text{g} \end{aligned} \quad (2)$$

where a is the partial specific volume of alkali lignin and b is the partial specific volume of DMSO at 25 °C (Sigma Aldrich, 2023; Martin et al., 1967; LeBel and Goring, 1962).

4. Find the maximum dry bulk density of the soil being used in the mix

Similar to ordinary Portland cement concrete, a higher fraction of soil per unit volume is correlated with greater compressive strength for BSC. In the case of soil-based materials, the dry bulk density is dependent on the solvent content, Sol_v_c , and compaction methods used to manufacture specimens. We used a double compaction method to manufacture test specimens according to a modified version of ASTM standard C29 (ASTM, C29 et al., 2009). Prior studies using the same aggregate and the same double compaction method have shown

Table 1
Design methodology for biopolymer-bound soil composites using lignin.

Step	Description	Parameter	Source
1	Find a solvent that can dissolve the biopolymer into solution	Dimethylsulfoxide (DMSO)	Experiments
2	Determine the maximum amount of biopolymer that can be dissolved into solution	$(\%w/w_{BP})_{HL} = 47\%$ $(\%w/w_{BP})_{AL} = 58\%$	Experiments Eq. (1)
3	Find the density of the biopolymer solution ρ_{BPsol}	$\rho_{BPsol}(47\%) = 1.18 \text{ g/cm}^3$ $\rho_{BPsol}(58\%) = 1.21 \text{ g/cm}^3$	Eq. (2)
4	Find the maximum dry bulk density of for the soil	$\rho_{dry,bulksoil}^{max} = 1.48 \text{ g/cm}^3$	Literature or experiments
5	Find the density of the soil particles	$\rho_{soil} = 2.60 \text{ g/cm}^3$	Experiments
6	Calculate the minimum volume fraction of voids V_v^{min}	$V_v^{min} = 0.43$	Eq. (3)
7	Determine the void saturation ratio	V_{BPsol}/V_v^{min}	
8	Determine the mass of soil, biopolymer, and solvent for desired volume of BSC	$m_{soil}, m_{BP}, m_{sol}$	Table 2

Table 2
Lignin-based biopolymer-bound soil composite design sheet.

Section I: Table Inputs & Constants								
Inputs	m_{soil}	m_{BP}	m_{DMSO}	m_{Total}	Constants	$\rho_{dry,bulksoil}^{max}$	ρ_{soil}	V_v^{min}
$\rightarrow$	I_1	I_2	I_3	$I_1 + I_2 + I_3$	$\rightarrow$	1.48 g/cm^3	2.60 g/cm^3	Eq. (3)
Calculate Outputs using: $\%w_{soil} = \frac{m_{soil}}{m_{Total}}(i)$, $\%w_{BP} = \frac{m_{BP}}{m_{Total}}(ii)$, $\%w_{DMSO} = \frac{m_{DMSO}}{m_{Total}}(iii)$								
$\%w_{BP/S} = \frac{\%w_{BP}}{\%w_{soil}}(iv)$, $solv_c = \frac{\%w_{DMSO}}{\%w_{soil}}(v)$, $\frac{V_{BPsol}}{V_v^{min}} = \frac{\%w_{BP/S}}{\%(w/w)_{BP}}(vi)$								
Section II: Table Outputs								
Outputs	$\%w_{soil}$	$\%w_{BP}$	$\%w_{DMSO}$	$\%w_{BP/S}$	ρ_{BPsol}	$\%(w/w)_{BP}$	$Solv_c$	$\frac{V_{BPsol}}{V_v^{min}}$
$\rightarrow$	use (i)	use (ii)	use (iii)	use (iv)	Eq. (2)	Eq. (1)	use (v)	use (vi)

that the maximum dry bulk density for Grade 90 sand, $\rho_{dry,bulksoil}^{max}$ is 1.48 g/cm^3 (Rosa et al., 2020).

5. Find the density of the soil particles

The density of the soil particles can be found on product information sheets, prior literature, or through laboratory experiments (ASTM D854). In this study, Grade 90 sand was used, which, according to prior studies, has a soil particle density, ρ_{soil} , of 2.60 g/cm^3 (Rosa et al., 2020).

6. Calculate the minimum volume fraction of voids V_v^{min}

To achieve a mix that has optimal strength and is efficient in biopolymer usage, the volume of the biopolymer solution, V_{BPsol} , must not exceed that of the voids, V_v , at the maximum bulk density. The minimum void fraction, V_v^{min} , refers to the void volume (V_v), at the maximum bulk density of the soil before the addition of biopolymer.

$$V_v^{min} = 1 - \frac{\rho_{dry,bulksoil}^{max}}{\rho_{soil}} \quad (3)$$

7. Determine the void saturation ratio

To achieve the most efficient design for a given concentration of biopolymer in solution, the volume of voids should be filled completely with biopolymer solution. Therefore the void saturation ratio, V_{BPsol}/V_v^{min} , is calculated for each mix, with the preference being to design mixes with void saturation ratios near 1.

8. Calculate the mass of soil, biopolymer, and solvent to manufacture desired amount of BSC

To manufacture the desired amount of BSC, the mass of soil, biopolymer, and solvent needed is found in Table 2, which depicts the design sheet for lignin-based BSC mixes. For a given total mass of BSC, the user inputs (I_1 , I_2 , I_3), shown by the yellow cells, are varied until the designed void saturation ratio (V_{BPsol}/V_v^{min}) and biopolymer concentration, $(\%w/w_{BP})_{max}$, are satisfied. The inputs of the table are used to calculate the outputs of the table. The output representing biopolymer-to-soil ratio ($\%w_{BP/S}$) is highlighted in green as it can be used in conjunction with the design relationships (shown later in Fig. 6) presented in this study to tailor lignin-based BSC for a target compressive strength.

3. Experimental methods and materials

3.1. Aggregate and solvent

Feldspathic grade 90 sand (# 90 Silver Sand), from P.W. Gillibrand, Simi Valley, California, was used for all experiments. The physical properties of this material, as reported by the manufacturer, were grain shape — sub-angular; hardness — 6.5 Mohs; moisture content — $< 0.1\%$; bulk density — 1409 kg/m^3 ; soil particle density — 2.60 g/cm^3 ; and uniformity coefficient 1.75–1.95. DMSO, reagent grade, was selected as the solvent and obtained from Fisher Scientific.

3.2. Overview of technical lignins

There are many types of lignin, including kraft lignin (lignoboost lignin and lignoforce lignin), alkali lignin (soda), steam expansion lignin, hydrolysis lignin, organosolv lignin, and lignosulfonate lignin. Fig. 1 shows the different types of biomass from which various lignins are derived, known production volumes per year, and current uses of each type of lignin. Alkali lignin and hydrolysis lignin were selected for this study. Kraft lignin and lignosulfonate lignin comprise more than 90% of global lignin production (Mun et al., 2021; Aro and Fatehi, 2017). While alkali lignin is sometimes referred as kraft lignin, they are not identical. Alkali lignin has a higher molecular weight, more aryl-glycerol groups, and more enol-ether linkages than kraft lignin. Alkali lignin also has an estimated 8%–14% phenylglycerol structure, which is much higher than that of kraft lignin (Zhao et al., 2019). Hydrolysis lignin is a byproduct of bioethanol production, unlike alkali lignin and kraft lignin, which are products from the paper and pulp industry. By selecting alkali lignin and hydrolysis lignin, we are choosing lignins from two different industrial sources with the desire to generalize the applicability of the design principle shown in this study.

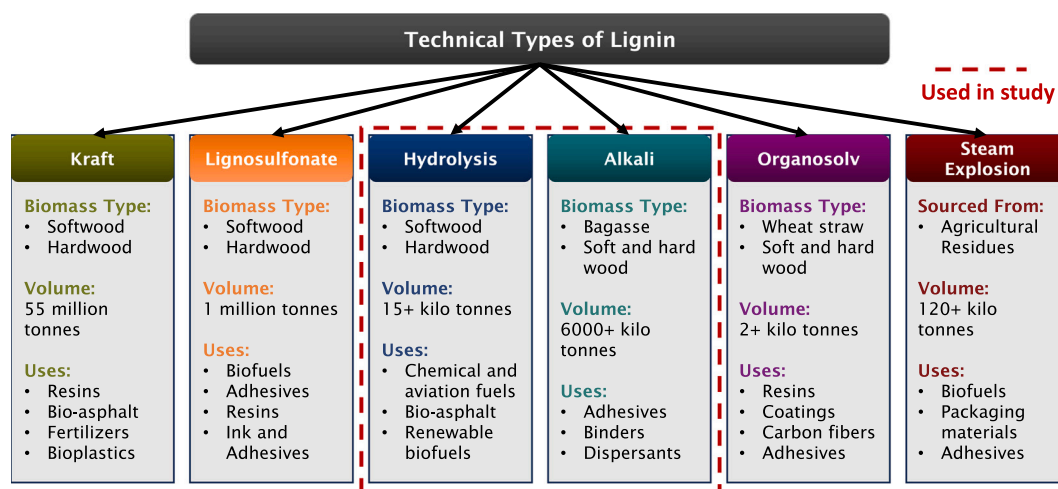


Fig. 1. Main types of technical lignins and associated information on biomass type, production volume, and current commercial applications (Mastrolitti et al., 2021; Bajwa et al., 2019). For this study, alkali lignin and hydrolysis lignin were investigated extensively to produce lignin-based BSC materials.

3.3. Lignin sources

Alkali lignin was obtained from Sigma-Aldrich (product #471003), with average molecular weight of 10,000 (Sigma Aldrich, 2023). Hydrolysis lignin was obtained from Shell International Exploration and Production, Inc., and produced by a mild acid pretreatment protocol and enzymatic hydrolysis for 2G bioethanol production, with an average molecular weight estimated to be 22,000 (Iram et al., 2022; Khazraie et al., 2017). No chemical modification (e.g., functionalization) was performed on either type of lignin prior to use for manufacturing lignin-based BSC.

3.4. Specimen manufacturing and testing

Before using hydrolysis lignin, the material was dry sieved using a 2 mm brass mesh sieve, to remove twigs and inorganic matter larger than 2 mm. For the alkali lignin, no sieve was needed. To make a lignin-based BSC mix, the lignin was first dissolved into DMSO by stirring with the aid of a KitchenAid stand mixer (Pro 500 Series 10). The process of dissolving the lignin was facilitated by the step-wise addition of DMSO. The process of dissolving the lignin took approximately 5–10 min.

After fully dissolving the lignin, the lignin solution and grade 90 sand were added to a mixing bowl. The stand mixer was used to blend the mixture for 5 min until the sand–DMSO–lignin mixture was uniform. A custom cylindrical molding apparatus measuring 13 mm inner diameter and 76 mm height, made from 316 stainless steel, was used to form the cylindrical shapes of the BSC specimens. A piston with a 12.8 mm diameter and 84 mm height was used to compact the specimens in the mold before the specimens were extruded and desiccation was begun. The compaction was performed using an MTS Criterion Model 43 electromechanical universal testing machine. Double 5 MPa compaction was implemented, which involves end-over-end compaction of each specimen (Rosa et al., 2020). After double 5 MPa compaction, cylindrical specimens were extruded from the mold by pushing on the piston. The height and weight of the specimens were measured before being placed in an oven (Fisher Scientific IsoTemp) at 60 °C until DMSO was completely removed. The degree of desiccation can be calculated by starting with the mass of DMSO in each specimen at the time of fabrication and tracking the mass loss over time. Complete desiccation of lignin-based BSC (i.e., 100% DMSO removal) takes approximately two weeks. After completion of desiccation, the specimens were sanded on both ends with the heights and masses again being measured before and after sanding. The heights and diameters of the specimens were measured at least 3 times using digital calipers that are accurate to the nearest 0.05 mm.

After being sanded, the specimens were subjected to unconfined uni-axial compression tests using an MTS Criterion Model 43 electromechanical universal testing machine, to determine ultimate compressive strength. The ultimate compressive strength was found by dividing the peak load (load at specimen failure) by the average cross-sectional area of the specimens. The tested specimens had roughly the same heights (± 1 mm) and diameters (± 0.50 mm). Specimens that did not fall within these dimensions were left out of the study. Fig. 2 shows a typical compressive strength testing specimen used in this study, along with a brick made from lignin-based BSC.

Table 3 shows the ten successful mixes that were prepared and tested during this study (five mix designs for each type of lignin). The mix name contains information regarding the type of aggregate (grade 90 sand), solvent (DMSO), biopolymer (alkali lignin or hydrolysis lignin), percent biopolymer in solution ($\%(w/w)_{BP}$), and biopolymer-to-soil ratio ($\%(w_{BP/s})$).

4. Experimental results

Results of unconfined uniaxial compression tests of lignin-based BSC specimens are shown in Table 4. For each mix, 8–10 specimens were prepared. Average compressive strength and standard deviation for these measurements are shown in Table 4. The relationships between mix parameters (void saturation ratio, solvent content, and biopolymer content) and experimentally obtained compressive strength and densities (unit weight and dry bulk soil density) are shown in Fig. 3 and in Fig. 4.

A simple, linear model appears to adequately account for the relationship between compressive strength and void saturation ratio, as shown in Fig. 3a. To fit the relationship between compressive strength and void saturation ratio, a zero point was added to the fit. This is because of the physical relationship between compressive strength and void saturation ratio, as specimens with near zero void saturation ratio will have strengths that are near zero (no biopolymer in the voids of the composite). The equation that describes this relationship for each type of lignin-based BSC is shown in Table 5 Eq. (4). A simple, linear model also appears to account for the relationship between compressive strength and biopolymer content (Fig. 3c). As per the void saturation ratio, a zero point was added to the relationship between compressive strength and biopolymer content, as BSC with near zero biopolymer (no biopolymer in the BSC) content will have a near zero compressive strength. Eq. (5) describes this relationship for each type of lignin-based BSC and is shown in Table 5.

Fig. 4a and Fig. 4b depict the relationship between unit weight and dry bulk soil density, respectively, with compressive strength for

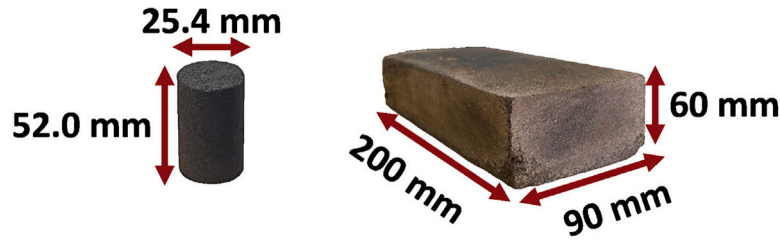


Fig. 2. Lignin-based BSC specimens used in this Study. The cylindrical specimen shown on the left is the type of specimen used for mechanical testing.

Table 3
Experimental mix designs of lignin-based biopolymer-bound soil composites.

Mix #	Mix name ^a	%w _{soil}	%w _{BP}	%w _{DMSO}	%w _{BP/s}	Sol v _c	%(w/w) _{BP}	$\frac{V_{BP,soil}}{V_{min}}$
Hydrolysis Lignin (HL)								
1	S90_DMSO_HL_BPS_37_BP_12	75.9%	9.0%	15.2%	11.8%	20.0%	37.1%	0.94
2	S90_DMSO_HL_BPS_38_BP_09	80.0%	7.5%	12.5%	9.4%	15.6%	37.5%	0.74
3	S90_DMSO_HL_BPS_41_BP_12	77.1%	9.4%	13.5%	12.2%	17.6%	40.9%	0.87
4	S90_DMSO_HL_BPS_42_BP_14	75.7%	10.3%	14.0%	13.6%	18.4%	42.4%	0.94
5	S90_DMSO_HL_BPS_47_BP_15	75.2%	11.5%	13.2%	15.4%	17.6%	46.5%	0.96
Alkali Lignin (AL)								
6	S90_DMSO_AL_BPS_50_BP_16	75.8%	12.1%	12.1%	16.0%	16.0%	50.0%	0.92
7	S90_DMSO_AL_BPS_50_BP_17	74.9%	12.6%	12.5%	16.9%	16.7%	50.2%	0.97
8	S90_DMSO_AL_BPS_52_BP_15	77.6%	11.8%	10.6%	15.1%	13.7%	52.4%	0.83
9	S90_DMSO_AL_BPS_54_BP_19	74.1%	14.1%	11.8%	19.0%	15.9%	54.5%	1.00
10	S90_DMSO_AL_BPS_58_BP_18	76.1%	13.8%	10.1%	18.2%	13.3%	57.8%	0.89

^a The mix names describe the aggregate_solvent_biopolymer_%(w/w)_{BP}_%w_{BP/s}.

Table 4
Design and compressive strength of lignin-based BSC mixes.

Mix #	Mix name	Fresh unit weight γ_{BSC}^{fresh} (g/cm ³)	Fresh bulk soil density $\rho_{bulksoil}^{fresh}$ (g/cm ³)	Unit weight γ_{BSC} (g/cm ³)	Dry bulk soil density $\rho_{bulksoil}^{dry}$ (g/cm ³)	Compressive strength σ_c (MPa)
Hydrolysis Lignin (HL)						
1	S90_DMSO_HL_BPS_37_BP_12	1.64 ± 0.07	1.25	1.51 ± 0.03	1.35	3.95 ± 0.78
2	S90_DMSO_HL_BPS_38_BP_09	1.52 ± 0.02	1.21	1.40 ± 0.02	1.28	1.59 ± 0.79
3	S90_DMSO_HL_BPS_41_BP_12	1.71 ± 0.04	1.32	1.49 ± 0.02	1.33	4.28 ± 0.75
4	S90_DMSO_HL_BPS_42_BP_14	1.79 ± 0.02	1.36	1.63 ± 0.03	1.43	6.91 ± 1.71
5	S90_DMSO_HL_BPS_47_BP_15	1.82 ± 0.02	1.36	1.62 ± 0.03	1.40	8.13 ± 1.94
Alkali Lignin (AL)						
6	S90_DMSO_AL_BPS_50_BP_16	1.69 ± 0.02	1.38	1.67 ± 0.01	1.44	5.21 ± 0.75
7	S90_DMSO_AL_BPS_50_BP_17	1.69 ± 0.02	1.38	1.72 ± 0.05	1.47	4.58 ± 0.74
8	S90_DMSO_AL_BPS_52_BP_15	1.58 ± 0.03	1.34	1.60 ± 0.03	1.39	3.62 ± 0.51
9	S90_DMSO_AL_BPS_54_BP_19	1.77 ± 0.03	1.44	1.80 ± 0.03	1.51	6.13 ± 0.55
10	S90_DMSO_AL_BPS_58_BP_18	1.82 ± 0.03	1.41	1.73 ± 0.04	1.46	6.79 ± 1.02

each type of lignin-based BSC. We chose not to develop a model relating unit weight and compressive strength, because unit weight is a variable that is based on both partial soil density and partial density of the biopolymer. Dry bulk soil density is a better choice because it is independent of biopolymer content. Unlike biopolymer content and void saturation ratio, which can be specified in the mix design, dry bulk soil density can only be determined after the fabrication of a BSC specimen. The dry bulk soil density, (partial soil density), g/cm³, is arrived at by determining the unit weight (specimen density), g/cm³, and subtracting the partial density of biopolymer, g/cm³. Lignin-based BSC with higher dry bulk soil density, exhibits higher compressive strength, as shown in Eq. (6). Physically, the dry bulk soil density is analogous to the degree of soil packing within the composite. Thus, BSCs with a higher dry bulk soil density will have soil particles that are more tightly packed together, and as a result, will have greater compressive strength. Fig. 3d depicts the relationship between biopolymer content and dry bulk soil density. For each type of lignin-based BSC, a higher amount of biopolymer content was correlated with a higher dry bulk soil density. This makes sense, physically, as there are potentially more inter-particle interactions/biopolymer bridges that form in BSC

that uses more biopolymer. Thus, we expect that BSC with a higher biopolymer content will have soil grains that are more tightly bound, resulting in greater dry bulk soil density.

Biopolymer content and dry bulk soil density were better correlated with compressive strength than was void saturation ratio (Fig. 3a) for both types of lignin-based BSC. We considered the possibility that a model based on both biopolymer content and dry bulk soil density might show an even better correlation with compressive strength than the univariate models. A linear regression using both biopolymer content and dry bulk soil density showed an overall marginal improvement over uni-variate models for lignin-based BSC. Since, there was not an obvious correlation between solvent content and compressive strength, we did not apply a regression model to the data in Fig. 3b.

Fig. 5 shows the overall performance of the four linear regression models shown in Table 5 for various lignin-based BSC mixes. Percent error was defined as the difference between the predicted compressive strength and the actual compressive strength for each experiment data point. Considering the combined data (for both hydrolysis lignin and alkali lignin), the four models showed similar ability to predict ultimate compressive strength, as judged by the R^2 values (as shown in Table 5).

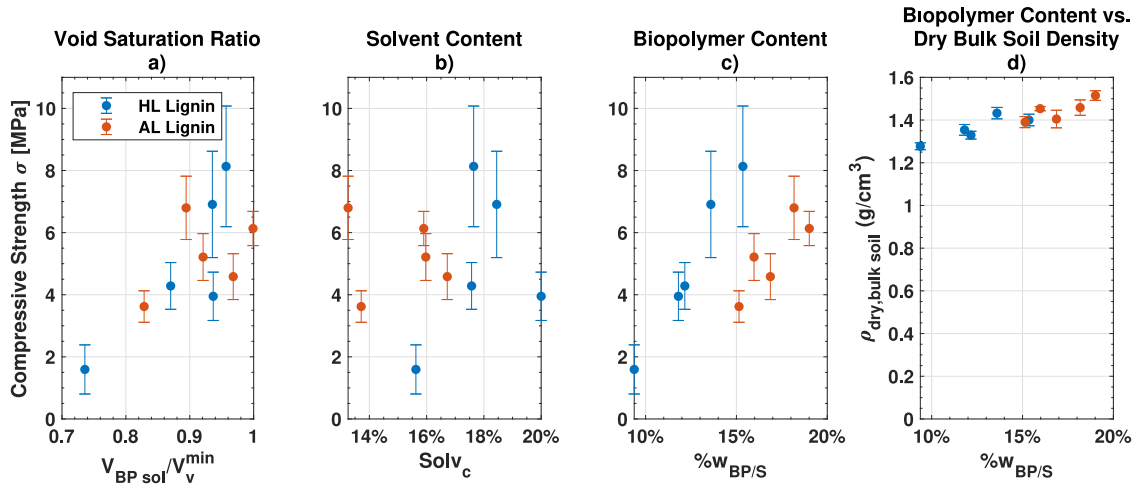


Fig. 3. Lignin-based BSC for specimen compressive strength against void saturation ratio, solvent content, and biopolymer content. The error bars correspond to standard deviation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

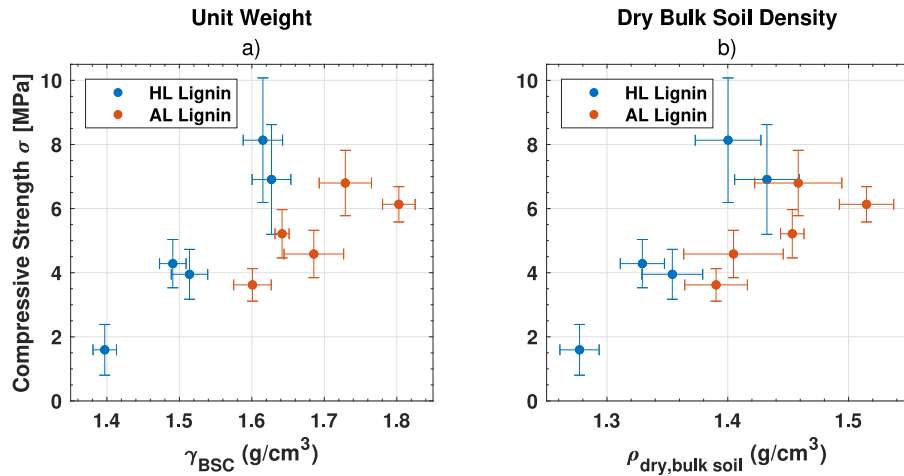


Fig. 4. Lignin-based BSC compressive strength as a function of a) unit weight, and b) dry bulk soil density. Both vertical and horizontal error bars correspond to standard deviation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5

Models for predicting the compressive strength of lignin-based BSCs.

Fitted Models	Equations and Fit Parameters for Predicting BSC Strength			
Model-1	Equation 4	$\sigma_c = a * \frac{V_{BP_{sol.}}}{V_v^{min}} + d$	(4)	
	Fitted parameters	$\sigma_{Hydrolysis} = [a = 6.5, d = -0.7]$ $\sigma_{Alkali} = [a = 5.8, d = -0.1]$		
Model-2	Equation 5	$\sigma_c = b * \%w_{BP/S} + d$	(5)	
	Fitted parameters	$\sigma_{Hydrolysis} = [b = 49.8, d = -1.0]$ $\sigma_{Alkali} = [b = 32.4, d = -0.2]$		
Model-3	Equation 6	$\sigma_c = c * \rho_{dry, bulk soil} + d$	(6)	
	Fitted parameters	$\sigma_{Hydrolysis} = [c = 39.1, d = -48.2]$ $\sigma_{Alkali} = [c = 20.3, d = -24.1]$		
Model-4	Equation 7	$\sigma_c = c * \rho_{dry, bulk soil} + b * \%w_{BP/S} + d$	(7)	
	Fitted parameters	$\sigma_{Hydrolysis} = [c = 10.4, b = 90.3, d = -20.4]$ $\sigma_{Alkali} = [c = 7.9, b = 47.5, d = -14.3]$		
Types of Lignin BSCs	Performance of Fitted Models for Predicting BSC Strength			
	Model-1	Model-2	Model-3	Model-4
	$R^2 = 0.61$	$R^2 = 0.78$	$R^2 = 0.84$	$R^2 = 0.99$
	$R^2 = 0.82$	$R^2 = 0.90$	$R^2 = 0.64$	$R^2 = 0.76$

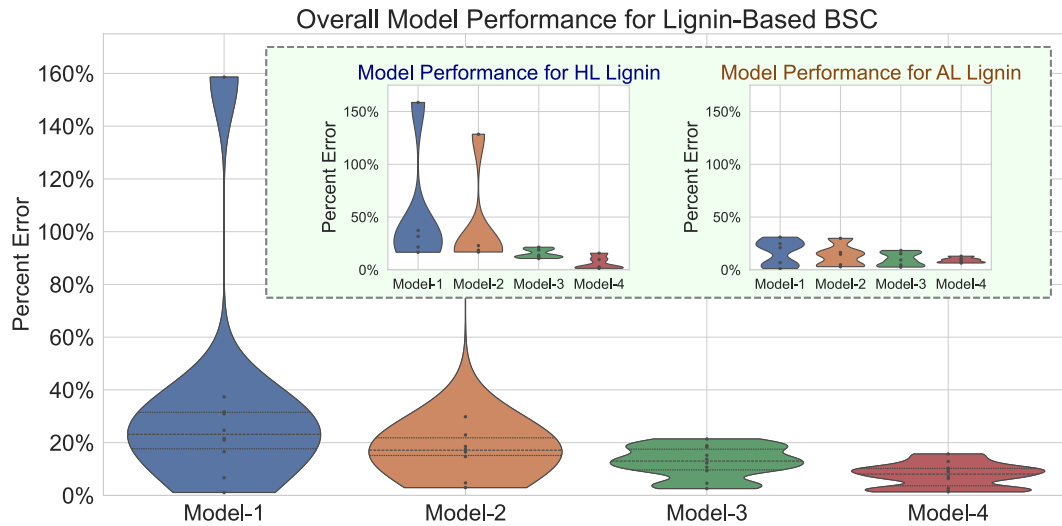


Fig. 5. Overall percent error of compressive strength of lignin-based BSC in comparison to predicted specimen strengths for different regression models from Eqs. (4)–(7) for all lignin mixes investigated in this study. The mini-window shows the model percent errors obtained for HL lignin-based BSC and AL lignin-based BSC.

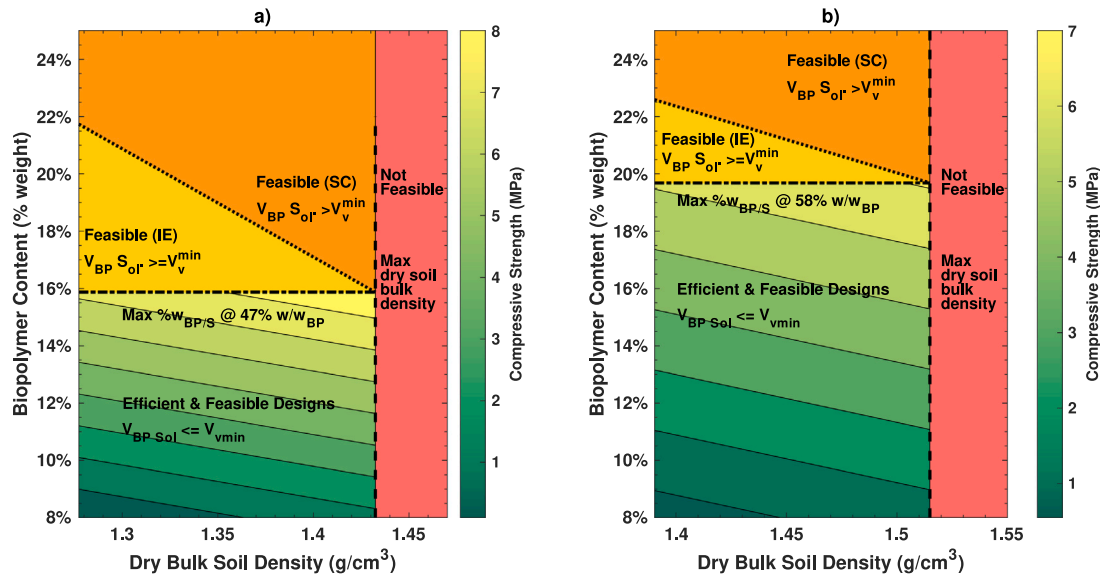


Fig. 6. Design relationships for (a) Hydrolysis Lignin-Based BSC and (b) Alkali Lignin-Based BSC. The yellow and orange regions show the feasible areas for design that are inefficient or more prone to shrinkage cracking respectively. The red region represents the infeasible area, based on the maximum dry bulk soil density of the aggregate. The contour region in the plot is the design area of the guide. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Model-4, which uses both dry bulk soil density and biopolymer-to-soil ratio (biopolymer content) as independent variables demonstrated slightly higher predictive ability.

We created contour plots to show the relationship between the variables dry bulk soil density and biopolymer-to-soil ratio, in relation to compressive strength. The contour regions of Fig. 6a and Fig. 6b represent the efficient and feasible design area (void saturation ratio is lower than 1). The yellowish-orange and orange regions of the design relationships depict the feasible but inefficient (IE) and feasible but subject to shrinkage cracking (SC) regions. Feasible but inefficient (IE) refers to mixes that are potentially feasible to make, but are inefficient since too much biopolymer solution has been added to the mixture. The feasible but subject to shrinkage cracking (SC) region shows mixtures that are inefficient and also more likely to experience shrinkage cracking. Finally, the red region of the figure depicts the infeasible region, as there are no specimens that can have a dry bulk soil density greater than the maximum dry bulk soil density.

5. Discussion of results

Our results show that for lignin-based BSC, hydrolysis lignin is a more effective biopolymer than alkali lignin from the standpoint of compressive strength. This is evident from a comparison of the results for mix #5 and mix #6, which had very similar biopolymer content. Specimens made from mix #5 had a compressive strength of 8.1 MPa and specimens made from mix #6 had a compressive strength of 5.2 MPa. The difference between hydrolysis lignin-based BSC and alkali lignin-based BSC, in terms of compressive strength, is also evident in Fig. 3c, which shows a steeper slope for hydrolysis lignin-based BSC versus alkali lignin-based BSC, as judged by the fitted parameters from the linear regression (Model-2, Eq. (5), in Table 5). The slope relating biopolymer content and compressive strength was 49.8 for hydrolysis lignin-based BSC and 32.4 for alkali lignin-based BSC. Although the difference in slope was not significant at the 95% confidence level, it was significant at the 75% confidence level (data not shown). Additional

data or tests exploring the mechanical properties of the biopolymer or interfacial bond strength between the biopolymer and aggregate, may help to confirm and determine the underlying reasons that lead to this apparent difference between hydrolysis lignin-based BSC and alkali lignin-based BSC.

Since dry bulk soil density is a predictor of BSC compressive strength, we considered the possibility that the higher compressive strength that we measured for hydrolysis lignin-based BSC versus alkali lignin-based BSC might be due to higher dry bulk soil density in the finished specimens of hydrolysis lignin-based BSC. However, we observed only a small difference in dry bulk soil density for the two types of lignin-based BSC, with the alkali lignin-based BSC specimens actually having the higher dry bulk soil density, not the hydrolysis lignin-based BSC specimens. We conclude that the higher compressive strength of hydrolysis lignin-based BSC is due to a fundamental difference between the two biopolymers, and not due to a difference in the resulting dry bulk soil density for the finished specimens. The greater strength of biopolymer bridges between sand particles or greater strength of the interface between the biopolymer and the sand particles for hydrolysis lignin versus alkali lignin likely accounts for the observed difference in compressive strength. Future studies involving simple tests performed on dried samples of lignin, by itself, may help to address this topic. While this study focused on the initial design and manufacture of lignin-based BSC, further investigations into the microstructure of lignin-based BSC are needed to better understand the interactions (pore-pore, soil-soil, and soil-biopolymer) between the phases of the composite (Rosa et al., 2016; Lou et al., 2023a).

The maximum strength for BSC is achieved when (1) the concentration of biopolymer dissolved into the solution is at maximum and (2) the voids of the BSC are fully filled (i.e., $\frac{V_{BP}^{sol}}{V_{min}} = 1$). This has been confirmed by Rosa et al. (Rosa et al., 2020). The maximum biopolymer concentration ($\%w/w_{BP}$) in DMSO with our given equipment was 47 % for hydrolysis lignin and 58 % for alkali lignin. A practical limitation to the maximum solubility of lignin in DMSO relates to the workability of the lignin solution, which can become quite viscous as the concentration of lignin approaches the solubility limit.

The design relationships shown in Fig. 6 were developed using the experimental results of this study and allow users to implement lignin-based BSC in non-structural construction applications. Figs. 3 and 4 show that the relationships between compressive strength and key mix parameters can be approximated as linear relationships within the limits of the data quality of this study.

Lignin is a naturally occurring biopolymer which is underutilized but has great potential for use in construction materials. The development of sustainable lignin-based BSC is significant for three reasons: (1) being the first application of a non-protein plant-derived biopolymer from a waste stream in BSC, (2) the first instance of use of a hydrophobic (non-water soluble) biopolymer in forming BSC, and (3) being a major step towards the development of a sustainable construction material. The development of BSC using a hydrophobic biopolymer is attractive as it indicates that this version of BSC will likely exhibit better resistance to water exposure without the need for protective coatings. Although the strength of lignin-based BSC is lower than previous BSC using bovine serum albumin or other blood-proteins, BSC made with lignin is still promising as it can be used in the construction industry for non-structural applications (e.g., pavement, non-load-bearing-walls, etc.).

Additionally, the development of lignin-based BSC provides three obvious benefits: (1) reduction of the high carbon footprint associated with the production of conventional construction materials, (2) a new way to utilize increasing amounts of lignin resulting from the growth of bioethanol production, and (3) reducing the emission of carbon from the combustion of lignin when it is not used for material production. Lignin-based BSC utilizes lignin as a biopolymer binder to provide strength in the composite material. In comparison to prior BSC made

with blood-proteins, lignin-based BSC provides better carbon sequestration due to the greater carbon richness of lignin (lignin ≈ 60 % carbon and bovine serum albumin ≈ 45 % carbon). Hence, the sequestration of carbon-rich lignin in lignin-based BSC is yet another benefit that helps make lignin-based BSC a sustainable alternative to conventional construction materials (Miao et al., 2024b).

Fig. 7 shows the range of density (unit weight) and compressive strength of lignin-based BSC in comparison to existing BSC and conventional materials (variants of lightweight concrete and ordinary portland cement concrete). Lignin-based BSC is less dense than conventional construction materials. By having a lightweight material, the dead load of lignin-based BSC will be lower than that of many materials, and allow users more flexibility in design. In comparison to other variants of BSC, lignin-based BSC has a similar compressive strength to BSC made from various gums (xanthan gum and guar gum) (Muguda et al., 2017), but a lower compressive strength compared to BSC made from animal blood plasma (AP920) (Rosa et al., 2020).

Currently, lightweight concrete (specifically, concrete made using lightweight aggregate) is used to produce a variety of lightweight structural members and non-structural building elements. These include pre-cast wall panels, interior partition walls, roofing tiles, and other types of facade applications (Mohammed and Hamad, 2014; Thienel et al., 2020). Any of these applications are viable paths for introducing lignin-based BSC into the construction industry.

Fig. 8 shows the general principles for design and manufacture of lignin-based BSC. Dry bulk soil density, although it cannot be directly specified in the manufacture of lignin-based BSC, shows a strong correlation with the compressive strength of the finished material. Minimizing excess biopolymer solution (to keep void saturation ratio ≤ 1) allows higher dry bulk soil density to be achieved in the finished material. Maximizing the dry bulk soil density maximizes the packing of the soil particles in BSC which improves the mechanical interaction between soil particles, contributing to higher compressive strength.

The manufacture of lignin-based BSC involves dissolving lignin into DMSO to create the biopolymer solution. Once the lignin is fully dissolved, the granular material (sand, in this study) is blended with the biopolymer solution to form the wet mix. Simple molds can be used to shape the wet mix.

Upon setting a target design strength, the biopolymer-to-soil ratio can be determined from Fig. 6. The next decision to be made is the volume of solvent to be used in the mix. The volume of solvent must be sufficient to fully dissolve the biopolymer, but an excess must be avoided to ensure that the volume of the biopolymer solution is kept at or below the volume of voids in the soil. Mixes that use a volume of biopolymer solution that exceeds the void space (i.e., void saturation ratio > 1) will tend to exhibit shrinkage cracking during desiccation. Furthermore, overfilling the predicted void space within the soil will decrease the dry bulk soil density of the finished material, which will, in turn, decrease the compressive strength of the BSC.

Future studies into lignin-based BSC should be conducted to investigate the use of other types of lignin, such as kraft lignin, organosolv lignin, and lignosulfonate lignin. By utilizing the methodology presented in this study, new design relationships for these additional types of lignin could be produced. Water soluble lignins are increasingly being investigated (Jeffri et al., 2024). For instance, lignin produced in the kraft pulping process is hydrophobic but can be rendered water-soluble through copolymerization with acrylic acid (propenoic acid) (Kong et al., 2015). These hydrophilic lignins are currently not widely available for scalable use but are promising, as BSC using a water-based solvent could potentially have a lower life cycle impact than BSC using organic solvents.

Finally, a worthwhile option to consider is the addition of cellulose fibers to the biocomposite. This option is worth pursuing as cellulose fibers would be expected to increase the tensile strength of the biocomposite. This is an attractive idea because lignin and cellulose fibers are typically produced in the same industrial processes.

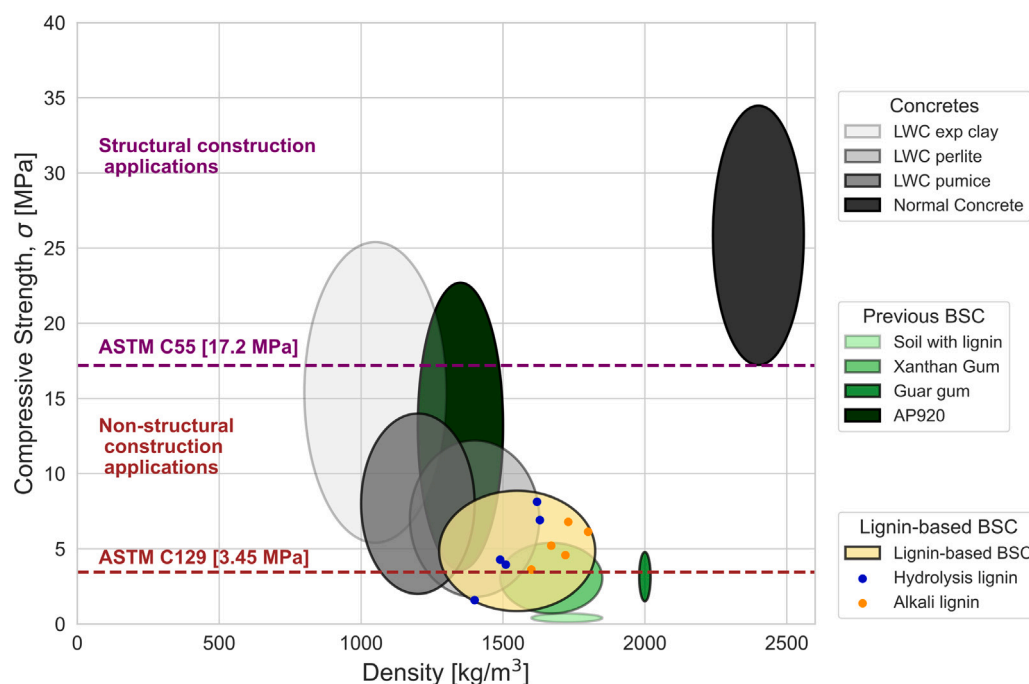


Fig. 7. Ashby plot comparing the compressive strength and density (unit weight) of lignin-based BSC from this study to other conventional materials and types of BSC. The family of gray ovals shows the approximate regions of unit weight and compressive strength for various forms of lightweight concrete (LWC) and ordinary portland cement concrete (OPCC). The family of green ovals shows unit weight and compressive strength regions for existing forms of BSC. The yellow oval represents the approximate region for lignin-based BSC, with the blue and orange data points representing the compressive strength and density results obtained in this study. Materials that meet the requirements of ASTM C129 can be used for non-structural construction applications (pavement/roads, roofing tiles, and non-structural elements), while materials that meet the requirements of ASTM C55 can be used for structural applications. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

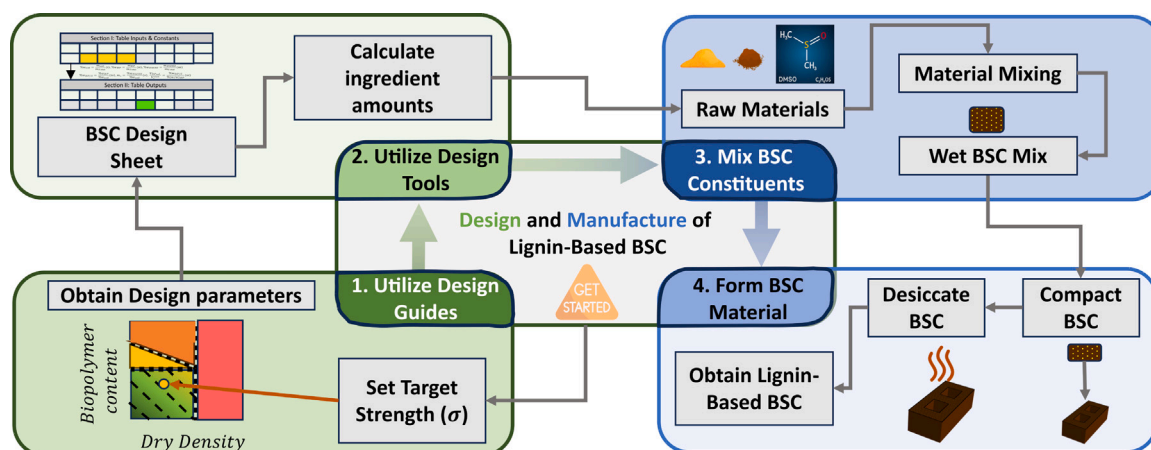


Fig. 8. Overall framework for the design and manufacture of lignin-based biopolymer-bound soil composites (BSC). The left portion of the chart shows the design procedure/materials available, while the right side shows the procedure for manufacturing lignin-based BSCs.

6. Conclusions

In this paper we introduce a new variant of BSC, lignin-based BSC, that uses lignin as a biopolymer binder. Two types of lignin — hydrolysis lignin and alkali lignin — were used in this study, and both biopolymers yielded viable biocomposites with similar compressive strength. Considering both types of lignin, we achieved compressive strength ranging from 1.6–8.1 MPa, which, according to ASTM C129, allows lignin-based BSC to be used for non-structural construction applications (e.g., pavement, roofing tiles, and non-structural elements), but not for structural construction applications (structural members where OPCC is used). Lignin-based BSC opens a significant new arena in the field of sustainable construction materials.

Our use of lignin in the development of this material represents the first use of a non-protein plant-derived biopolymer to facilitate

fabrication of a cement-free granular construction material. Unlike existing variants of BSC, which use hydrophilic biopolymers, the use of a hydrophobic biopolymer in this variant of BSC allows it to be implemented in a wider range of environments (locations more prone to humid weather or rainfall). Prior to this study, BSC used hydrophilic biopolymer binders, which would require the use of protective coatings if they were to be deployed in wet locations. Additionally, the biopolymers used to manufacture lignin-based BSC are sourced from the waste streams of major industries (paper/pulp production and biofuels production facilities), and are independent of the fossil fuel industry. While proteins used in the manufacture of BSC have the advantage of carbon sequestration, this advantage is even greater for BSC made with lignin, as lignin is more rich in carbon than most proteins.

From this study, we found that hydrolysis lignin was a more efficient binder as compared to alkali lignin for manufacture of lignin-based BSC. Further study of these biocomposites should investigate the interfacial bond strength between lignin and the granular material and should investigate the strength of the biopolymer itself in the dried state. The introduction of lignin-based BSC represents a significant step towards developing a cement-free granular construction material and will help foster a green transition in the construction of the built environment.

CRedit authorship contribution statement

Barney H. Miao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Robert J. Headrick:** Writing – review & editing, Resources. **Zhiye Li:** Writing – review & editing, Resources. **Leonardo Spanu:** Resources. **David J. Loftus:** Writing – review & editing, Conceptualization. **Michael D. Lepech:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by Shell International Exploration and Production, Inc., through its membership in the Stanford Strategic Energy Alliance. The lignin material in this study was obtained from Shell collaborators. Their support of this work is gratefully acknowledged. Facility support for this work was provided by the John A. Blume Earthquake Engineering Center at Stanford University. The authors would like to acknowledge Andrew Lesh, Yinjian Li, Peter Harijanto, and Kimberly Wong for their experimental support. The authors would also like to acknowledge Kyle Douglas for assistance in material acquisition and laboratory support.

Appendix A

Sample Calculation for Dry Bulk Soil Density of Lignin-Based BSC

For this sample calculation, we will be referring to Mix #1 shown in Tables 3 and 4. The dry bulk soil density (partial density of soil) is found by subtracting the partial density of the biopolymer from the unit weight (specimen density), g/cm³, of the lignin-based BSC.

$$\rho_{dry, bulk soil} = \gamma_{BSC} - \rho_{biopolymer}$$

After initial preparation, before desiccation takes place, Mix #1 consists of 75.9% sand, 9.0% lignin, and 15.2% DMSO. After desiccation (after loss of all DMSO), Mix #1 consists of 89.4% sand and 10.6% lignin, by weight.

$$\rho_{dry, bulk soil} = 1.51 \text{ g/cm}^3 - (0.106)(1.71 \text{ g/cm}^3) = 1.35 \text{ g/cm}^3$$

Alternatively, according to Rosa et al. we can also find the dry bulk soil density using the following equation (Rosa et al., 2020):

$$\rho_{dry, bulk soil} = \frac{\gamma_{BSC}}{1 + \%w_{BP/s}}$$

$$\rho_{dry, bulk soil} = \frac{1.51 \text{ g/cm}^3}{1 + 0.118} = 1.35 \text{ g/cm}^3$$

Data availability

Data will be made available on request.

References

- Albuhairi, D., Di Sarno, L., 2022. Low-carbon self-healing concrete: state-of-the-art, challenges and opportunities. *Buildings* 12 (8), 1196.
- Aro, T., Fatehi, P., 2017. Production and application of lignosulfonates and sulfonated lignin. *ChemSusChem* 10 (9), 1861–1877.
- ASTM, C29, et al., 2009. Standard Test Method for Bulk Density (“Unit Weight”) and Voids in Aggregate. American Society for Testing and Materials, Annual Book: West Conshohocken, PA, USA.
- Ayeldeen, M.K., Negm, A.M., El Sawwaf, M.A., 2016. Evaluating the physical characteristics of biopolymer/soil mixtures. *Arab. J. Geosci.* 9, 1–13.
- Bajwa, D., Pourhashem, G., Ullah, A.H., Bajwa, S., 2019. A concise review of current lignin production, applications, products and their environmental impact. *Ind. Crop. Prod.* 139, 111526.
- Boarino, A., Klok, H.-A., 2023. Opportunities and challenges for lignin valorization in food packaging, antimicrobial, and agricultural applications. *Biomacromolecules* 24 (3), 1065–1077.
- Bribián, I.Z., Capilla, A.V., Usón, A.A., 2011. Life cycle assessment of building materials: Comparative analysis of energy and environmental impacts and evaluation of the eco-efficiency improvement potential. *Build. Environ.* 46 (5), 1133–1140.
- Brownsey, G.J., Noel, T.R., Parker, R., Ring, S.G., 2003. The glass transition behavior of the globular protein bovine serum albumin. *Biophys. J.* 85 (6), 3943–3950.
- Brunow, G., Lundquist, K., 2010. Functional groups and bonding patterns in lignin (including the lignin-carbohydrate complexes). *Lignin Lignans Adv. Chem.* 1.
- Canakci, H., Aziz, A., Celik, F., 2015. Soil stabilization of clay with lignin, rice husk powder and ash. *Geomech. Eng.* 8 (1), 67–79.
- Chang, I., Cho, G.-C., 2014. Geotechnical behavior of a beta-1, 3/1, 6-glucan biopolymer-treated residual soil. *Geomech. Eng.* 7 (6), 633–647.
- Chang, I., Im, J., Prasidhi, A.K., Cho, G.-C., 2015a. Effects of Xanthan gum biopolymer on soil strengthening. *Constr. Build. Mater.* 74, 65–72.
- Chang, I., Jeon, M., Cho, G.-C., 2015b. Application of microbial biopolymers as an alternative construction binder for earth buildings in underdeveloped countries. *Int. J. Polym. Sci.* 2015, 1–9.
- Collins, M.N., Nechifor, M., Tanasă, F., Zănoagă, M., McLoughlin, A., Stróżyk, M.A., Culebras, M., Teacă, C.-A., 2019. Valorization of lignin in polymer and composite systems for advanced engineering applications—a review. *Int. J. Biol. Macromol.* 131, 828–849.
- Dastpak, A., Lourencon, T.V., Balakshin, M., Hashmi, S.F., Lundström, M., Wilson, B.P., 2020. Solubility study of lignin in industrial organic solvents and investigation of electrochemical properties of spray-coated solutions. *Ind. Crop. Prod.* 148, 112310.
- De Schepper, M., Van den Heede, P., Van Driessche, I., De Belie, N., 2014. Life cycle assessment of completely recyclable concrete. *Materials* 7 (8), 6010–6027.
- Demuner, I.F., Gomes, F.J.B., Gomes, J.S., Coura, M.R., Borges, F.P., Carvalho, A.M.M.L., Silva, C.M., 2021. Improving kraft pulp mill sustainability by lignosulfonates production from processes residues. *J. Clean. Prod.* 317, 128286.
- Dove, C.A., Bradley, F.F., Patwardhan, S.V., 2016. Seaweed biopolymers as additives for unfired clay bricks. *Mater. Struct.* 49, 4463–4482.
- Figueiredo, P., Lintinen, K., Hirvonen, J.T., Kostianen, M.A., Santos, H.A., 2018. Properties and chemical modifications of lignin: Towards lignin-based nanomaterials for biomedical applications. *Prog. Mater. Sci.* 93, 233–269.
- Grabber, J.H., 2005. How do lignin composition, structure, and cross-linking affect degradability? A review of cell wall model studies. *Crop Sci.* 45 (3), 820–831.
- Hasanbeigi, A., Price, L., Lin, E., 2012. Emerging energy-efficiency and CO₂ emission-reduction technologies for cement and concrete production: A technical review. *Renew. Sustain. Energy Rev.* 16 (8), 6220–6238.
- Iram, A., Cekmecelioglu, D., Demirci, A., 2022. Integrating 1G with 2G bioethanol production by using distillers’ dried grains with solubles (DDGS) as the feedstock for lignocellulolytic enzyme production. *Fermentation* 8 (12), 705.
- Jeffri, N.I., Rawi, N.F.M., Kassim, M.H.B.M., Abdullah, C.K., 2024. Unlocking the potential: Evolving role of technical lignin in diverse applications and overcoming challenges. *Int. J. Biol. Macromol.* 133506.
- Katahira, R., Elder, T.J., Beckham, G.T., 2018. A brief introduction to lignin structure. *Khatami, H.R., O’Kelly, B.C., 2013. Improving mechanical properties of sand using biopolymers. J. Geotech. Geoenviron. Eng.* 139 (8), 1402–1406.
- Khazraie, T., Zhang, Y., Tarasov, D., Gao, W., Price, J., DeMartini, N., Hupa, L., Fatehi, P., 2017. A process for producing lignin and volatile compounds from hydrolysis liquor. *Biotechnol. Biofuels* 10, 1–14.
- Kong, X., Song, S., Wang, M., Zhao, Q., 2019. Experimental research of low liquid limit silt stabilized by lignin in the flooding area of yellow river. *Geotech. Geol. Eng.* 37, 5211–5217.
- Kong, F., Wang, S., Price, J.T., Konduri, M.K., Fatehi, P., 2015. Water soluble kraft lignin–acrylic acid copolymer: Synthesis and characterization. *Green Chem.* 17 (8), 4355–4366.
- Latifi, N., Horpibulsuk, S., Meehan, C.L., Abd Majid, M.Z., Tahir, M.M., Mohamad, E.T., 2017. Improvement of problematic soils with biopolymer—an environmentally friendly soil stabilizer. *J. Mater. Civ. Eng.* 29 (2), 04016204.

- LeBel, R., Goring, D., 1962. Density, viscosity, refractive index, and hygroscopicity of mixtures of water and dimethyl sulfoxide. *J. Chem. Eng. Data* 7 (1), 100–101.
- Liu, Y., Chang, M., Wang, Q., Wang, Y., Liu, J., Cao, C., Zheng, W., Bao, Y., Rocchi, I., 2020. Use of sulfur-free lignin as a novel soil additive: A multi-scale experimental investigation. *Eng. Geol.* 269, 105551.
- Lou, Z., Zhang, Y., Li, Y., Xu, L., 2023a. Study on microscopic physical and chemical properties of biomass materials by AFM. *J. Mater. Res. Technol.* 24, 10005–10026.
- Lou, Z., Zheng, Z., Yan, N., Jiang, X., Zhang, X., Chen, S., Xu, R., Liu, C., Xu, L., 2023b. Modification and application of bamboo-based materials: a review—Part II: application of bamboo-based materials. *Forests* 14 (11), 2266.
- Lu, Y.-C., Lu, Y., Fan, X., 2020. Structure and characteristics of lignin. *Lignin: Biosynthesis and Transformation for Industrial Applications*. Springer, pp. 17–75.
- Luo, H., Abu-Omar, M.M., 2017. Chemicals from lignin. In: *Encyclopedia of Sustainable Technologies*, vol. 3, Elsevier Amsterdam, The Netherlands, pp. 573–585.
- Mankar, A.R., Modak, A., Pant, K.K., 2022. Recent advances in the valorization of lignin: a key focus on pretreatment, characterization, and catalytic depolymerization strategies for future biorefineries. *Adv. Sustain. Syst.* 6 (3), 2100299.
- Martin, D., Weise, A., Niclas, H.-J., 1967. The solvent dimethyl sulfoxide. *Angew. Chem., Int. Ed. Engl.* 6 (4), 318–334.
- Mastrolitti, S., Borsella, E., Giuliano, A., Petrone, M., De Bari, I., Gosselink, R., van Erven, G., Annevelink, E., Traintafyllidis, K., Stichnothe, H., 2021. Sustainable Lignin Valorization: Technical Lignin, Processes and Market Development. Tech. Rep., IEA Bioenergy Task 42 & LignoCOST.
- Miao, B.H., Dong, Y., Theissler, A., Lesh, A.C., Loftus, D.J., Lepech, M.D., 2024a. Biosys: efficient quality control system for manufacturing of sustainable biopolymer composites. In: *Proceedings of the 11th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*. ACM, Association for Computing Machinery, Hangzhou, China, pp. 11–21.
- Miao, B.H., Headrick, R., Li, Z., Spanu, L., Loftus, D.J., Lepech, M.D., 2024b. Life cycle assessment and design of LignoBlock: A lignin bound block on the path towards a green transition of the construction industry. *J. Clean. Prod.* 143610.
- Mohammed, J.H., Hamad, A.J., 2014. Materials, properties and application review of lightweight concrete. *Tech. Rev. Fac. Eng. Univ. Zulia* 37 (2), 10–15.
- Muguda, S., Booth, S.J., Hughes, P.N., Augarde, C.E., Perlot, C., Bruno, A.W., Gallipoli, D., 2017. Mechanical properties of biopolymer-stabilised soil-based construction materials. *Géotech. Lett.* 7 (4), 309–314.
- Mun, J.-S., Pe, III, J.A., Mun, S.-P., 2021. Chemical characterization of kraft lignin prepared from mixed hardwoods. *Molecules* 26 (16), 4861.
- Pflieger, M., Kroflič, A., 2017. Acute toxicity of emerging atmospheric pollutants from wood lignin due to biomass burning. *J. Hazard. Mater.* 338, 132–139.
- Qureshi, M.U., Chang, I., Al-Sadarani, K., 2017. Strength and durability characteristics of biopolymer-treated desert sand. *Geomech. Eng.* 12 (5), 785–801.
- Roedel, H., 2017. On the Compressive Strength and Sustainability of Protein Bound Concretes (Ph.D. thesis). Stanford University.
- Rosa, I., Roedel, H., Allende, M.I., Lepech, M.D., Loftus, D.J., 2020. On designing biopolymer-bound soil composites (BSC) for peak compressive strength. *J. Renew. Mater.* 8 (8), 845.
- Rosa, I., Roedel, H., Lepech, M., Loftus, D., Garboczi, E., 2016. Three-phase statistically equivalent periodic unit cells for protein-bound soils. In: *15th Biennial ASCE Conference on Engineering, Science, Construction, and Operations in Challenging Environments*. American Society of Civil Engineers Reston, VA, pp. 775–786.
- Sameni, J., Krigstin, S., Sain, M., 2017. Solubility of lignin and acetylated lignin in organic solvents. *BioResources* 12 (1), 1548–1565.
- Sigma Aldrich, 2023. 471003 Alkali Lignin. <https://www.sigmaaldrich.com/US/en/product/aldrich/471003>. (Accessed 09 April 2023).
- Thienel, K.-C., Haller, T., Beuntner, N., 2020. Lightweight concrete—From basics to innovations. *Materials* 13 (5), 1120.
- Wang, Q., Han, H., Lou, Z., Han, X., Wang, X., Li, Y., 2022. Surface property enhancement of bamboo by inorganic materials coating with extended functional applications. *Composites A* 155, 106848.
- Zhang, T., Cai, G., Liu, S., 2018. Application of lignin-stabilized silty soil in highway subgrade: A macroscale laboratory study. *J. Mater. Civ. Eng.* 30 (4), 04018034.
- Zhang, T., Liu, S., Zhan, H., Ma, C., Cai, G., 2020. Durability of silty soil stabilized with recycled lignin for sustainable engineering materials. *J. Clean. Prod.* 248, 119293.
- Zhao, C., Huang, J., Yang, L., Yue, F., Lu, F., 2019. Revealing structural differences between alkaline and kraft lignins by HSQC NMR. *Ind. Eng. Chem. Res.* 58 (14), 5707–5714.